

ECE 361E: Machine Learning and Data Analytics for Edge AI

HW4 Assigned: Mar 3 DUE: Mar 12 (11:59:59pm CST)

Work in groups of two students. At the end of the PDF file, insert a paragraph where you describe each member's contribution and two valuable things you learned from this homework.

Only one submission per group is required.

Problem 1 [40p]: Structured Pruning

Question 1: [10p] Using the provided *main.py* file from the *HW4_files* directory, train the MobileNet-v1 model on CIFAR10. Fill in the *TODO* parts from the *prune_model.py* file to prune the MobileNet-v1 model using [Torch-Pruning](#). Prune **all layers** from MobileNet-v1 **except the last linear layer** using the *L1-norm* structured pruning technique. Run experiments with *fine-tuning epochs=5* and *pruning iteration=5* for the following pruning fractions *pruning_fraction=[0.05, 0.1, 0.2, 0.3, 0.4, 0.5]*.

Question 2: [15p] Re-run **all** experiments from **Problem 1, Question 1** with a new version of the *prune_network()* function that prunes **all layers** from MobileNet-v1 **except the first convolution layer, the first convolution block, and the last linear layer** using the *L1-norm* structured pruning technique. Draw two plots: one plot for the test accuracy vs. *pruning_fraction* and another plot for test accuracy vs. number of parameters. For both plots, start with the original (i.e., unpruned) MobileNet-v1 (i.e., *pruning_fraction=0*). Each plot will have two curves: one curve for the **Problem 1, Question 1** results and another curve for the **Problem 1, Question 2** results. Which pruning strategy works better? Explain.

Question 3: [15p] For the best pruning strategy only, deploy the pruned models with *pruning_fraction=[0.05, 0.1, 0.2, 0.3, 0.4, 0.5]*, along with the unpruned MobileNet-v1 (i.e., *pruning_fraction=0*), **only on Raspberry Pi 3B+** with ONNX and perform inference using the entire test set (available on the device in */home/student/HW4_files/test_deployment*); then complete *Table 1*:

Table 1

Pruning fraction	#Fine-tuning epochs	#parameters	Model Size [MB]	Maximum memory usage [MB]	Average latency <i>per</i> image [ms]	Average energy consumption <i>per</i> image [mJ]	Test accuracy [%]

Problem 2 [25p]: Static Post-Training Quantization

Question 1: [5p] Fill in the *TODO* parts from the *quantize_model.py* file to use static post-training quantization (PTQ) to convert the floating-point MobileNet-v1 model to an 8-bit INT ONNX model. Use this to quantize the unpruned MobileNet-v1.

Question 2: [5p] Deploy the ONNX quantized model on the Raspberry Pi 3B+ device and then complete *Table 2*:

Table 2

Pruning fraction	#Fine-tuning epochs	#parameters	Model Size [MB]	Maximum memory usage [MB]	Average latency <i>per</i> image [ms]	Average energy consumption <i>per</i> image [mJ]	Test accuracy [%]

Question 3: [15p] Compare and contrast the optimization of MobileNet-v1 using pruning and quantization. Based on your results from **Problem 1** and **Problem 2**, what are the pros and cons of using pruning alone and quantization alone in terms of latency, energy, accuracy, number of parameters and model size? Where does pruning excel in terms of optimization compared to quantization? Similarly, where does quantization excel in terms of optimization compared to pruning?

Problem 3 [35p+10Bp]: Inference Optimization

Question 1: [20p] Use all 6 pruned models obtained in **Problem 1, Question 3**, and on top of pruning, apply static post-training quantization (PTQ). Deploy the ONNX **pruned and quantized** models *only* on the Raspberry Pi 3B+ device and then complete Table 3:

Table 3

Pruning fraction	#Fine-tuning epochs	#parameters	Model Size [MB]	Maximum memory usage [MB]	Average latency <i>per</i> image [ms]	Average energy consumption <i>per</i> image [mJ]	Test accuracy [%]

Question 2: [15p] What are the advantages/disadvantages when using the pruning alone (**Table 1** from **Problem 1**), the quantization alone (**Table 2** from **Problem 2**), and the pruning + quantization together (**Table 3** from **Problem 3**)? Explain.

BONUS Question 3: [10Bp] Use any other types of pruning in combination with any types of quantization to obtain better results in terms of average latency, memory usage, or energy consumption, while keeping a similar accuracy performance. Add your new results in a new row at the end of **Table 3**. Explain your decisions for pruning, quantization, and their combination. Explain the results you obtain.

Submission Instructions

Include your solutions into a single zip file named <Team#>.zip. The zip file should contain:

1. A single PDF file containing all your results and discussions.
2. A *readme.txt* file explaining all your items in the zip file.
3. Your code files, named suggestively (e.g., **p1_q1.py** for **Problem 1 Question 1** code).

Good luck!